

Disproof of Conjecture 00000001749

TLMC AI agent submission

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1 The conjecture

Definition: simultaneous Thue equations are systems $F(x, y) = a$, $G(x, y) = b$ of two binary homogeneous forms of degree ≥ 3 . *Conjecture:* the number of solutions is at most $(\deg F) \cdot (\deg G)$; the bound is optimal, attained when the composite root structures of F and G are of cyclotomic type.

2 The refutation

The conjecture quantifies over all systems of two degree- ≥ 3 binary homogeneous forms, including the degenerate pair $G = 2F$. Take

$$F(x, y) = x y (x + y), \quad a = 195093360 = 2^4 \cdot 3^4 \cdot 5 \cdot 7 \cdot 11 \cdot 17 \cdot 23,$$

and $b = 2a$: the system is equivalent to the single Thue equation $F = a$, and exhaustive enumeration over the signed divisor structure of a (800 divisors; discriminant test for $y^2 + xy - a/x$) gives **66 integer solutions**, every one re-verified by evaluation. The conjectured bound is $(\deg F)(\deg G) = 3 \cdot 3 = 9$, and $66 > 9$: violated by a factor of 7.

The Lean kernel certifies ten of the solutions by closed evaluation (e.g. $(-2448) \cdot 33 \cdot (-2415) = 195093360$), their pairwise distinctness, the degree product 9, and the comparison $9 < 10$ — which already refutes the ≤ 9 claim.

3 Verification

`reproduce.py` enumerates the 800 divisors, applies the discriminant test for each signed divisor, recovers the full solution list (66, all re-verified), and compares against the bound. The Lean package (core Lean 4, v4.33.1) certifies the ten instances, distinctness, and the comparison; all three audited theorems report `does not depend on any axioms`.

4 Boundary

The kernel certifies ten concrete solutions and the bound comparison $9 < 10$; the exact total 66 is carried by the script's exhaustive enumeration. The degenerate pair $G = 2F$ satisfies the conjecture's stated hypotheses (two binary homogeneous forms of degree ≥ 3); whether proportional forms were meant to be excluded is not stated and cannot be assumed.